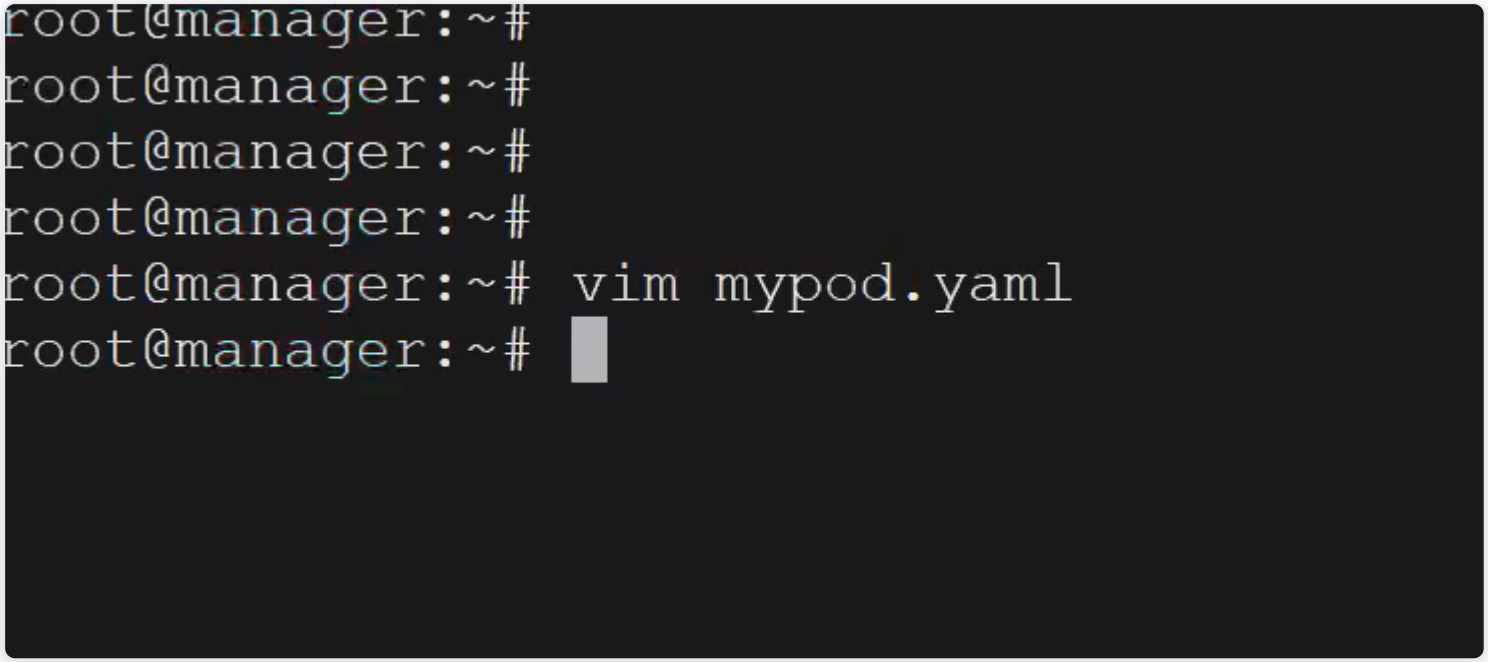
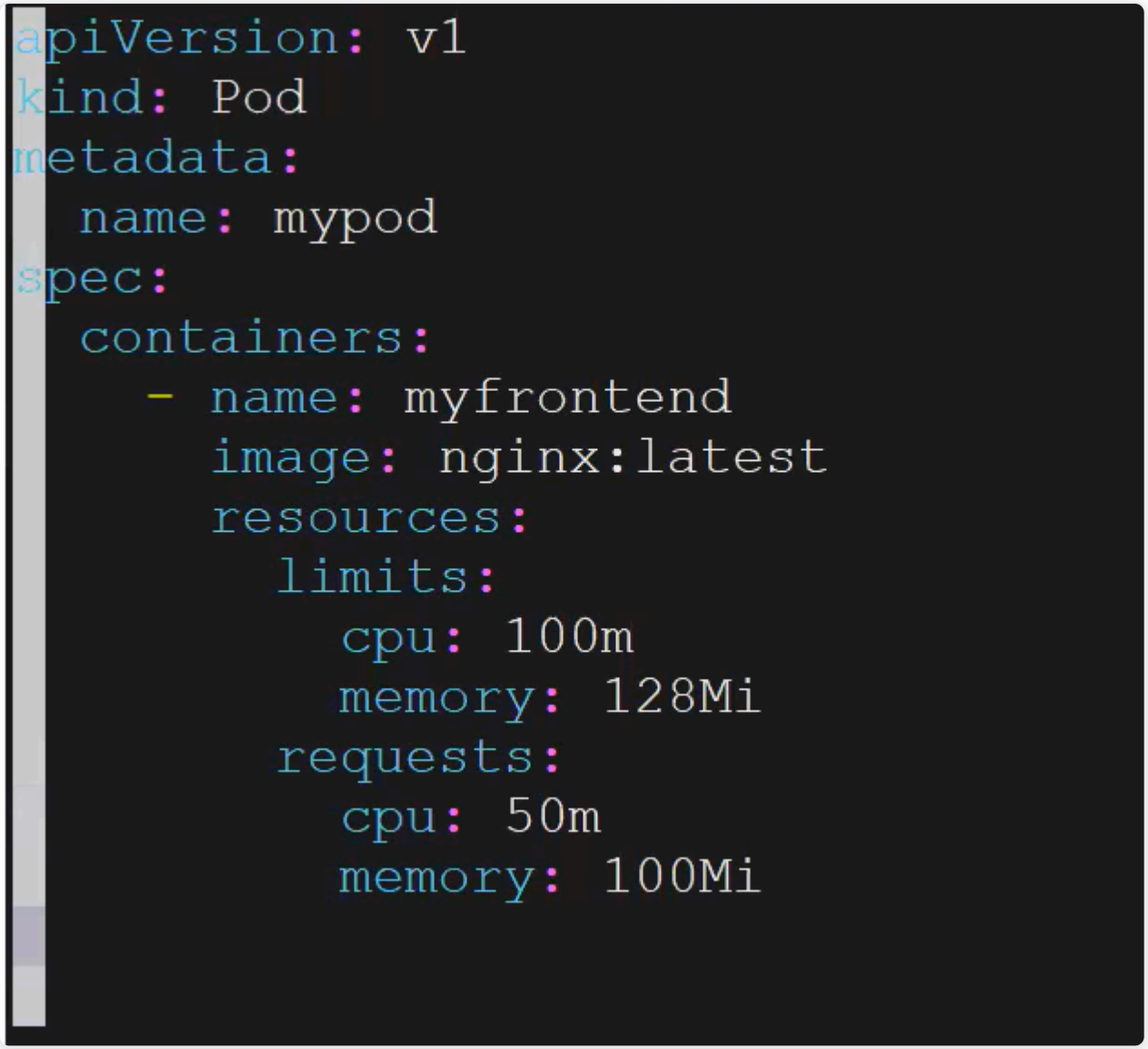
KUBERNETES CKA PREP › MY LEARNING NOTES › SESSION 04

Screenshots: ReplicaSet, Deployment & Service

Raw screenshots captured while watching this session, kept alongside the written notes (04-replicaset-deployment-and-service.md) for visual reference during interview revision.

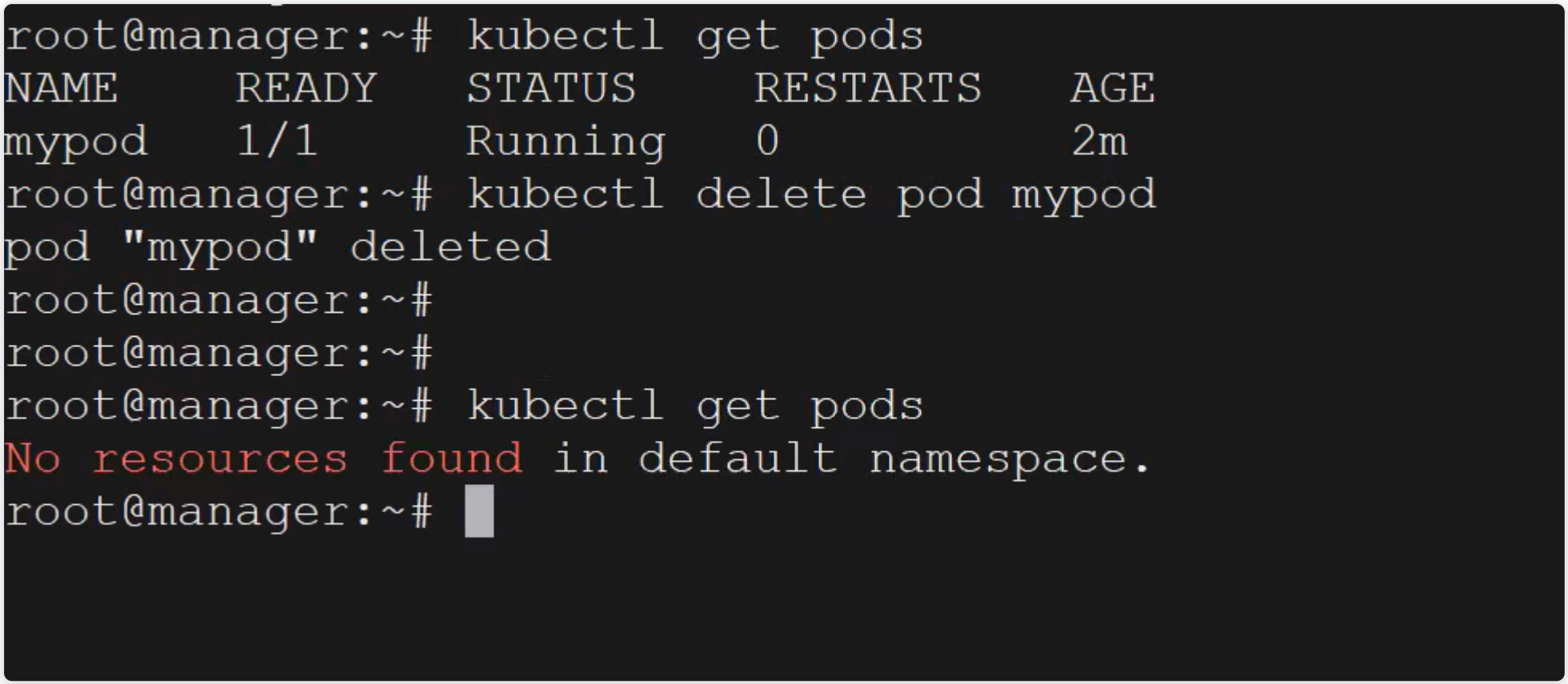
27 screenshots · captured 2026-09-11, 12:17 PM - 3:23 PM


root@manager:~#
root@manager:~#
root@manager:~#
root@manager:~#
root@manager:~# vim mypod.yaml
root@manager:~#



```yaml
apiVersion: v1
kind: Pod
metadata:
  name: mypod
spec:
  containers:
    - name: myfrontend
      image: nginx:latest
      resources:
        limits:
          cpu: 100m
          memory: 128Mi
        requests:
          cpu: 50m
          memory: 100Mi
```

```
root@manager:~# kubectl get pods
NAME    READY   STATUS    RESTARTS   AGE
mypod   1/1     Running   0          99s
root@manager:~#
root@manager:~#
root@manager:~# kubectl get pods -o wide
NAME    READY   STATUS    RESTARTS   AGE    IP              NODE      NOMINATED NODE   READ
INESS GATES
mypod   1/1     Running   0          104s   172.16.3.165    nodetwo   <none>           <non
e>
```

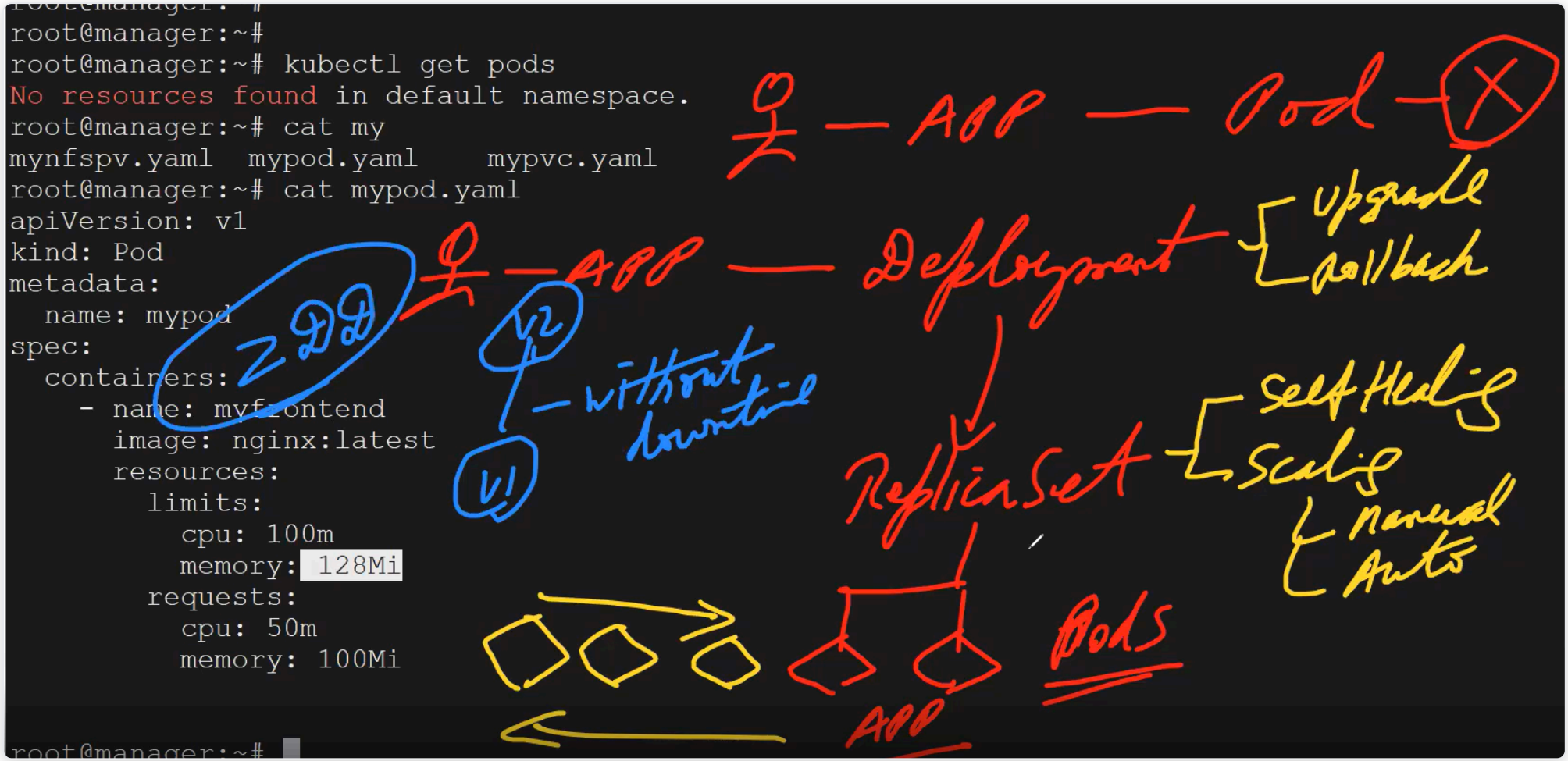

root@manager:~# kubectl get pods
NAME READY STATUS RESTARTS AGE
mypod 1/1 Running 0 2m
root@manager:~# kubectl delete pod mypod
pod "mypod" deleted
root@manager:~#
root@manager:~#
root@manager:~# kubectl get pods
No resources found in default namespace.
root@manager:~#

```
root@manager:~#
root@manager:~#
root@manager:~# kubectl get pods
No resources found in default namespace.
root@manager:~# cat my
mynfspv.yaml  mypod.yaml    mypvc.yaml
root@manager:~# cat mypod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: mypod
spec:
  containers:
    - name: myfrontend
      image: nginx:latest
      resources:
        limits:
          cpu: 100m
          memory: 128Mi
        requests:
          cpu: 50m
          memory: 100Mi
```



App — Pod [Singleton]

- No self Healing
- No Scaling
- Upgrade
- Roll back


root@manager:~#
root@manager:~# kubectl get pods
No resources found in default namespace.
root@manager:~# cat my
mynfspv.yaml mypod.yaml mypvc.yaml
root@manager:~# cat mypod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: mypod
spec:
  containers:
    - name: myfrontend
      image: nginx:latest
      resources:
        limits:
          cpu: 100m
          memory: 128Mi
        requests:
          cpu: 50m
          memory: 100Mi
root@manager:~#
App — Pod
App — Deployment
Upgrade
Rollback
v2
v1
without downtime
ReplicaSet
Self Healing
Scaling
Manual
Auto
Pods
App

Briefly summarize your current role and how you divide your time between project delivery and operations/support work. *

Describe your experience transitioning projects into steady-state operations after go-live. *

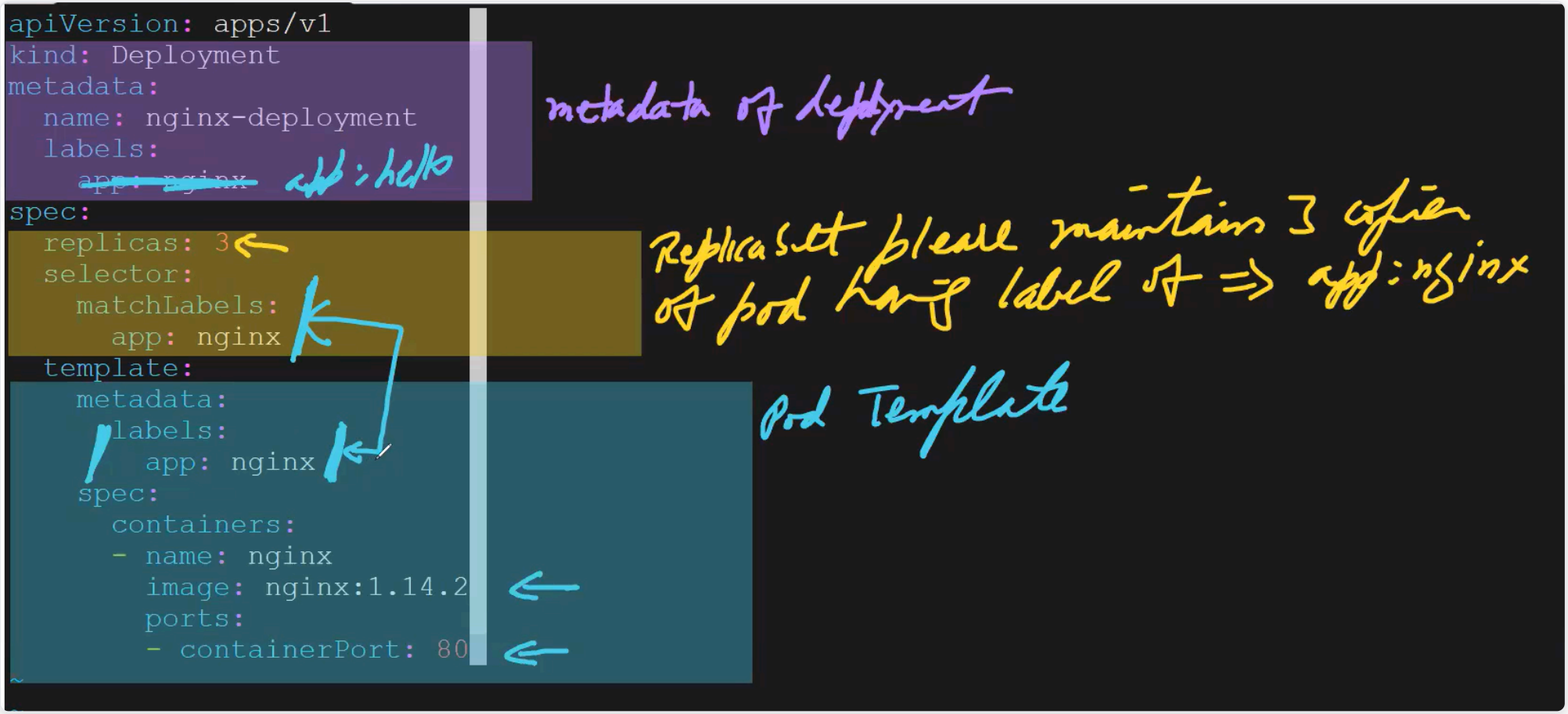

onments.*

State how many years you've worked in L2/L3 production support environments. *

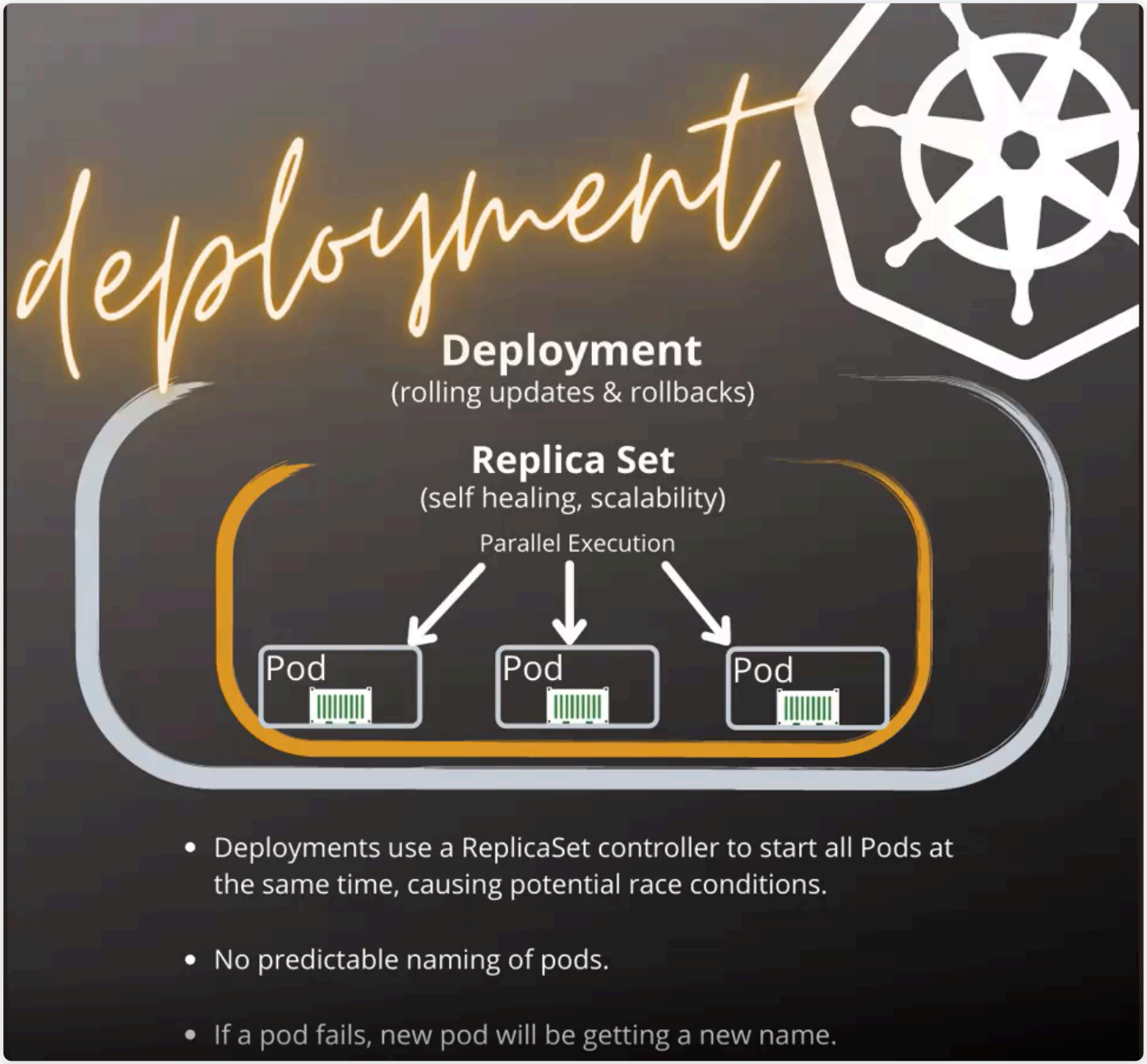
Outline your experience leading a 24×7 support model and managing multi-shift teams. *

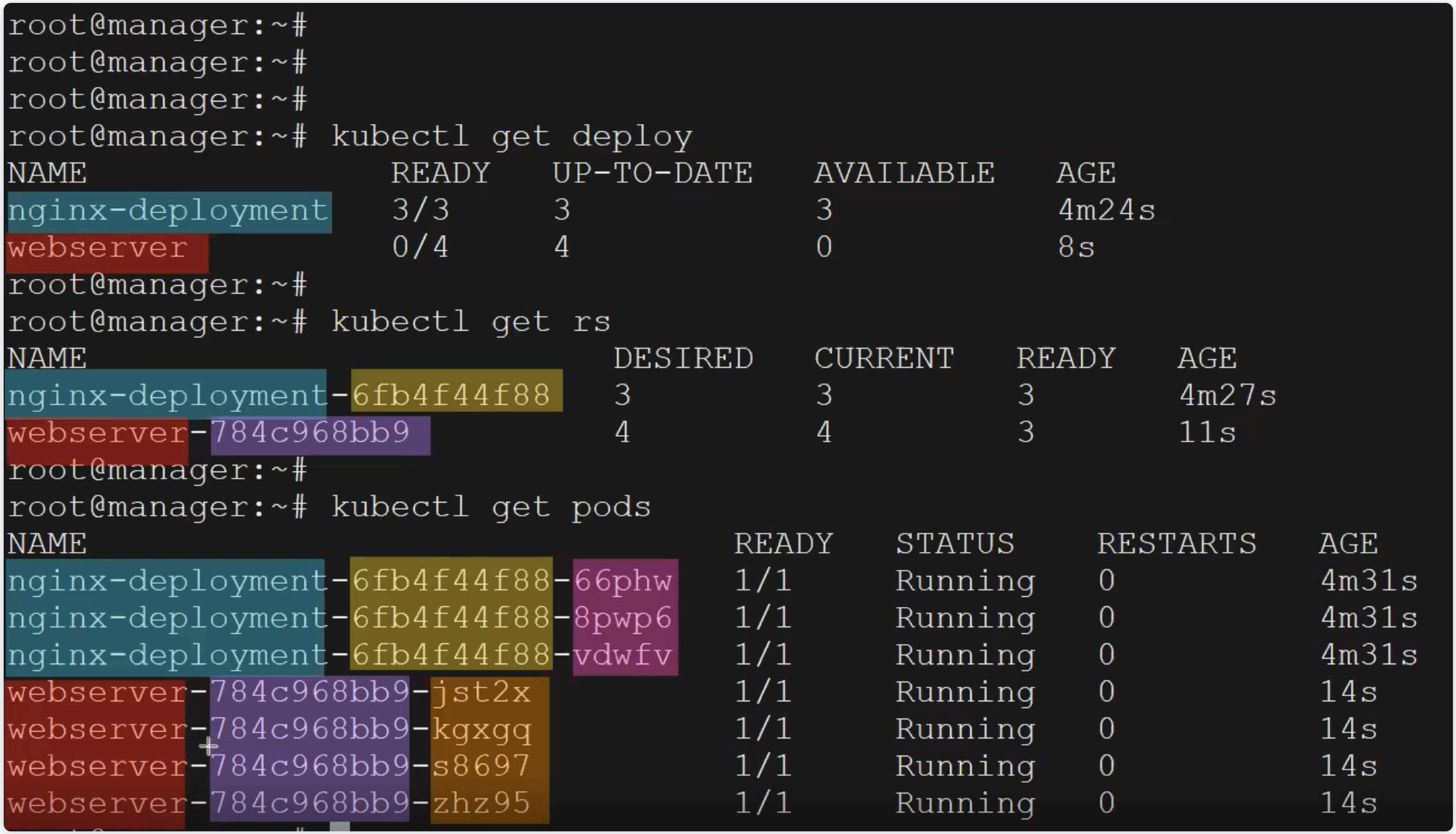
Share your experience handling high-priority incidents or escalations with enterprise customers. *

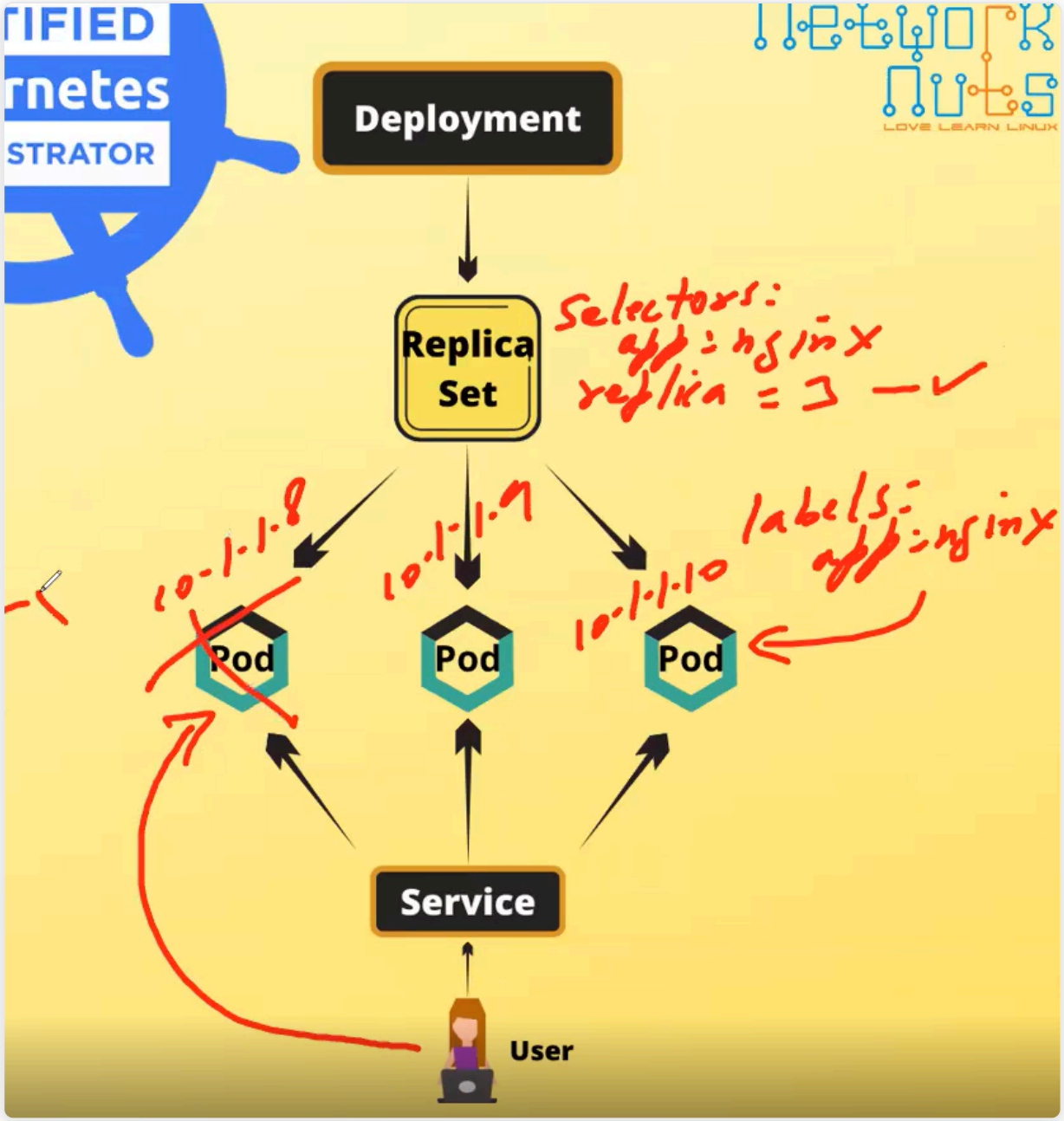
Describe your comfort and prior experience with on-call duties and weekend production support. *

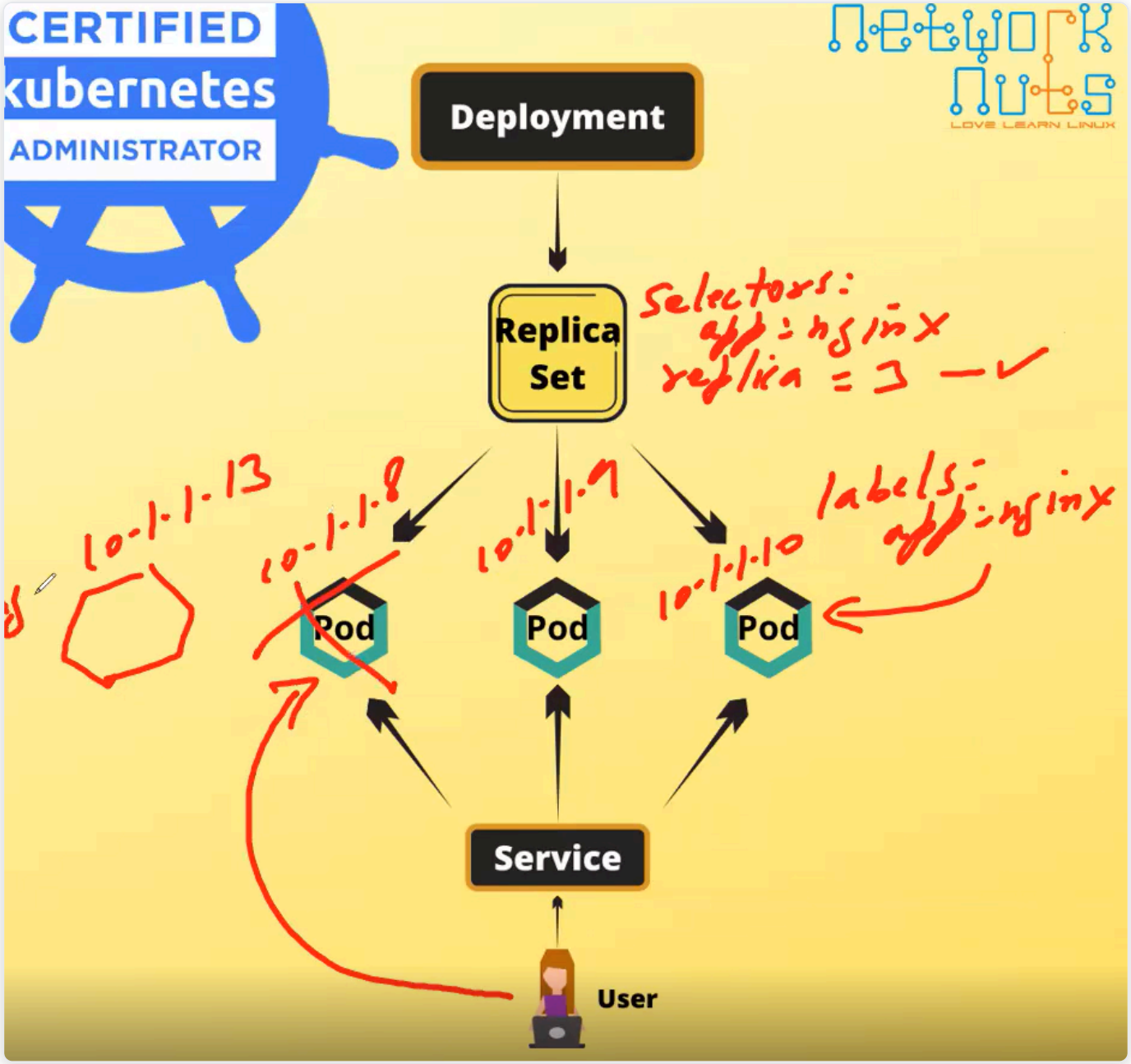

apiVersion: apps/v1
kind: Deployment
metadata:
name: nginx-deployment
labels:
app: nginx
app: hello
spec:
replicas: 3
selector:
matchLabels:
app: nginx
template:
metadata:
labels:
app: nginx
spec:
containers:
- name: nginx
image: nginx:1.14.2
ports:
- containerPort: 80
metadata of deployment
ReplicaSet please maintains 3 copies
of pod having label of => app: nginx
Pod Template

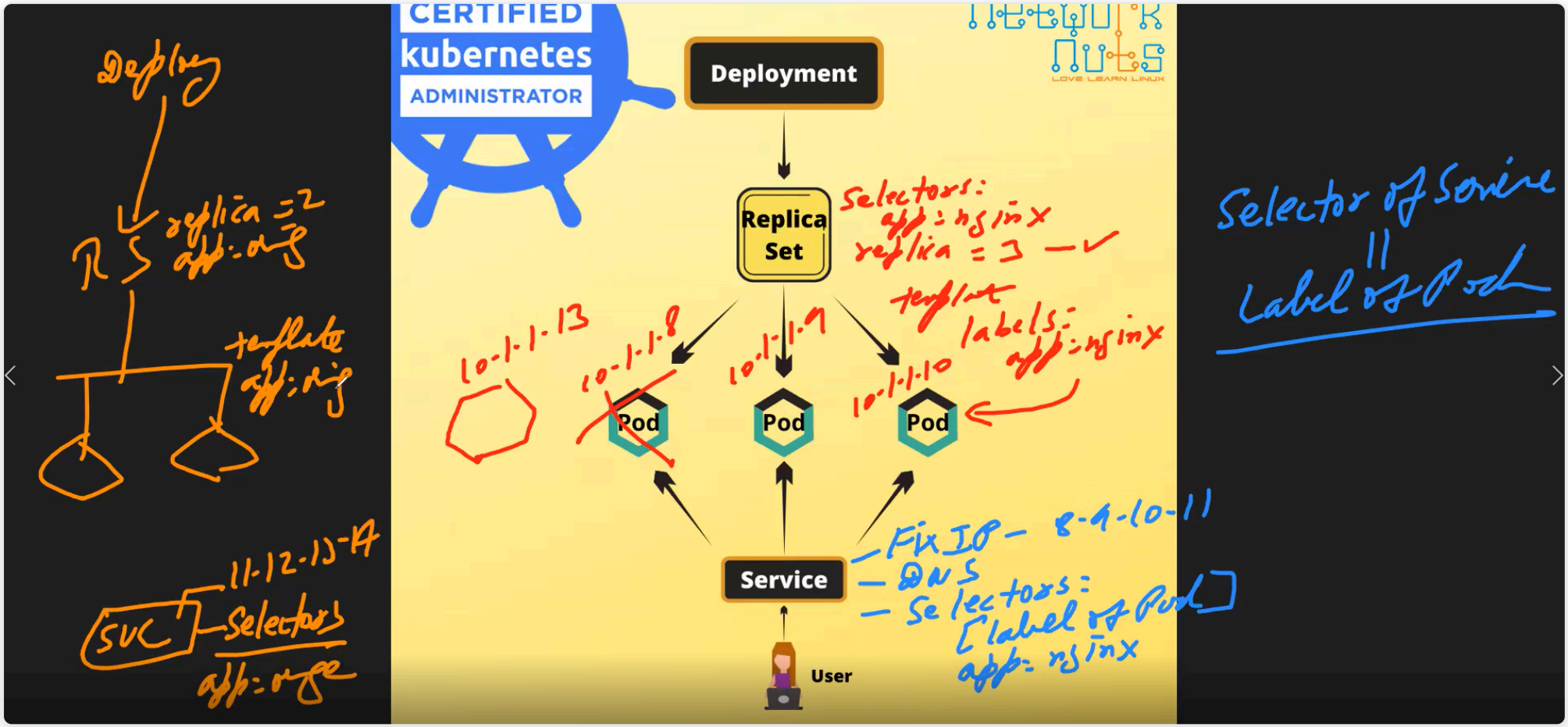

app: nginx
spec:
replicas: 3
selector:
matchLabels:
app: nginx
app: hello
template:
metadata:
labels:
app: nginx
spec:
containers:
- name: nginx
image: nginx:1.14.2
ports:
- containerPort: 80
resources:
limits:
cpu: 100m
memory: 100Mi
requests:
cpu: 50m
memory: 50Mi
metadata of deployment
ReplicaSet please maintain 3 copies of pod having label of ⇒ app: nginx
Pod Template

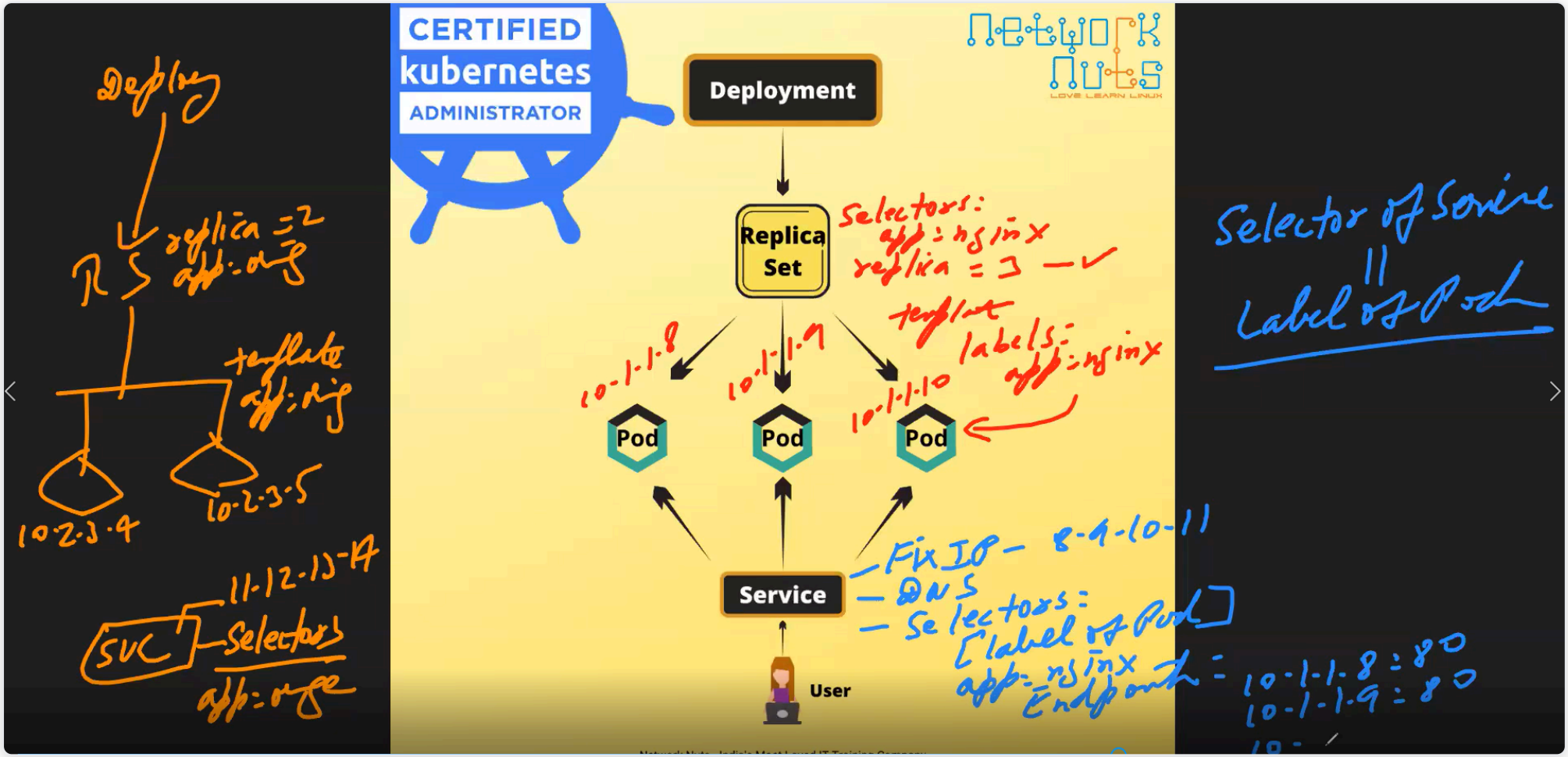

deployment
Deployment
(rolling updates & rollbacks)
Replica Set
(self healing, scalability)
Parallel Execution
Pod
Pod
Pod
• Deployments use a ReplicaSet controller to start all Pods at the same time, causing potential race conditions.
• No predictable naming of pods.
• If a pod fails, new pod will be getting a new name.

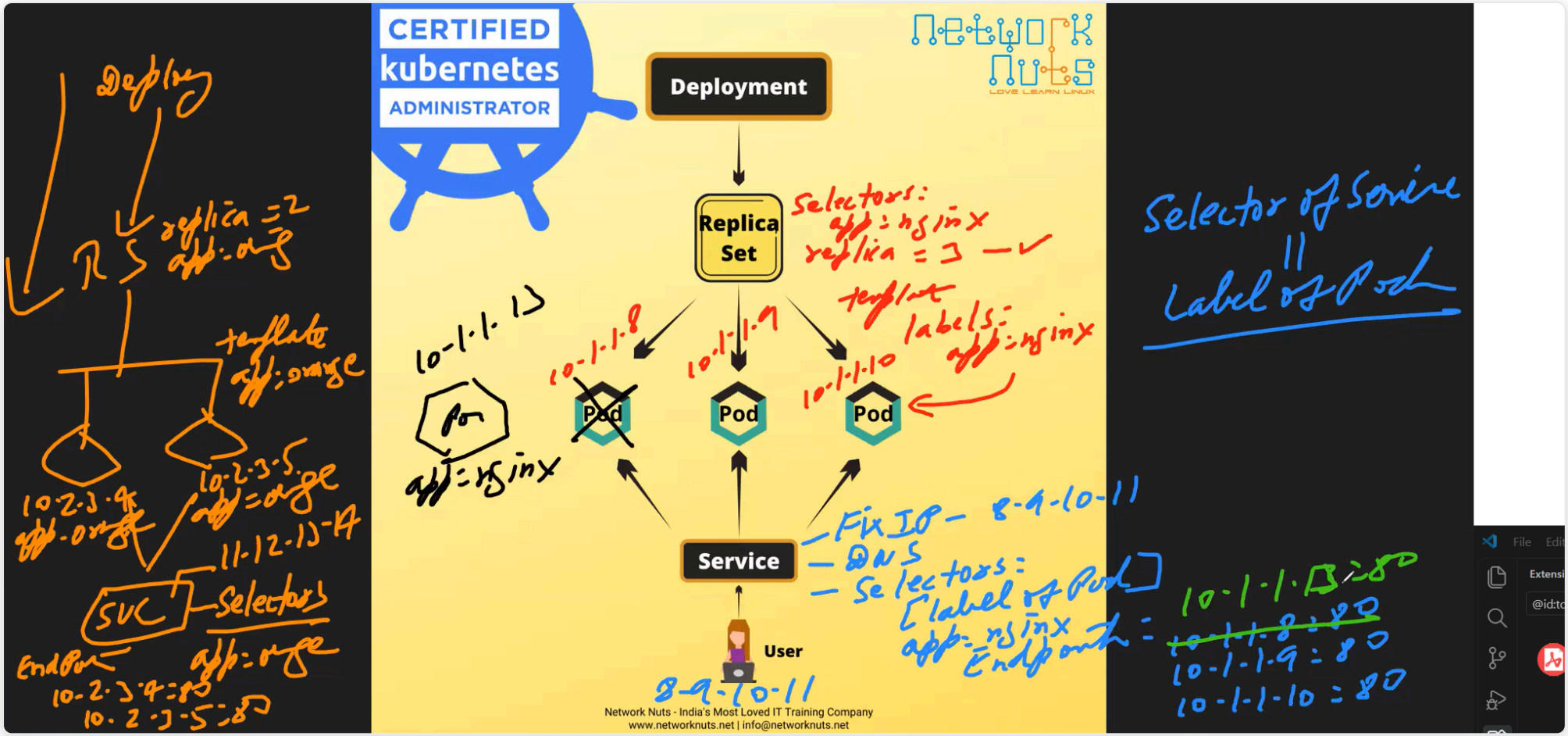

root@manager:~#
root@manager:~#
root@manager:~#
root@manager:~# kubectl get deploy
NAME READY UP-TO-DATE AVAILABLE AGE
nginx-deployment 3/3 3 3 4m24s
webserver 0/4 4 0 8s
root@manager:~#
root@manager:~# kubectl get rs
NAME DESIRED CURRENT READY AGE
nginx-deployment-6fb4f44f88 3 3 3 4m27s
webserver-784c968bb9 4 4 3 11s
root@manager:~#
root@manager:~# kubectl get pods
NAME READY STATUS RESTARTS AGE
nginx-deployment-6fb4f44f88-66phw 1/1 Running 0 4m31s
nginx-deployment-6fb4f44f88-8pwp6 1/1 Running 0 4m31s
nginx-deployment-6fb4f44f88-vdwfv 1/1 Running 0 4m31s
webserver-784c968bb9-jst2x 1/1 Running 0 14s
webserver-784c968bb9-kgxgq 1/1 Running 0 14s
webserver-784c968bb9-s8697 1/1 Running 0 14s
webserver-784c968bb9-zhz95 1/1 Running 0 14s

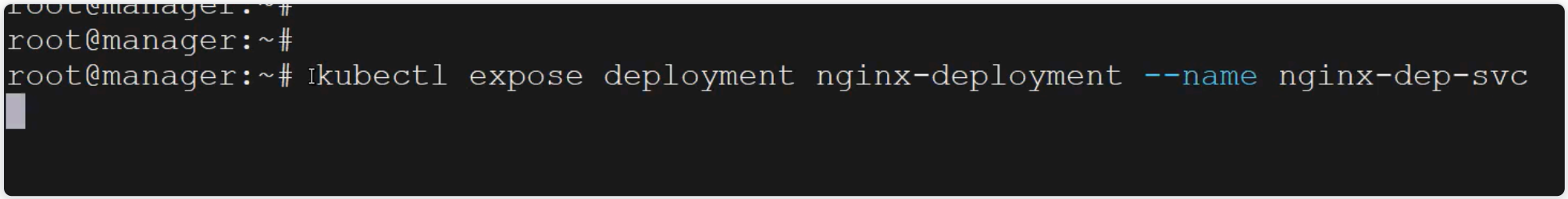

TIFIED
rnetes
STRATOR
Deployment
Replica Set
Selectors:
app: nginx
replica = 3
10.1.1.8
10.1.1.9
10.1.1.10
labels:
app: nginx
Pod
Pod
Pod
Service
User


CERTIFIED
kubernetes
ADMINISTRATOR
Deployment
Replica Set
Selectors:
app: nginx
replica = 3
10.1.1.13
10.1.1.8
10.1.1.9
10.1.1.10
labels:
app: nginx
Pod
Pod
Pod
Service
User


CERTIFIED
kubernetes
ADMINISTRATOR
Deployment
Replica Set
Selectors:
app: nginx
replica = 3
template
labels:
app: nginx
10.1.1.13
10.1.1.8
10.1.1.9
10.1.1.10
Pod
Pod
Pod
Service
User
Fix IP - 8.1.10.11
DNS
Selectors:
[label of Pod]
app: nginx
Deploy
RS
replica = 2
app: nginx
template
app: nginx
11.12.13.14
SVC
Selectors
app: nginx
Selector of Service
=
Label of Pod

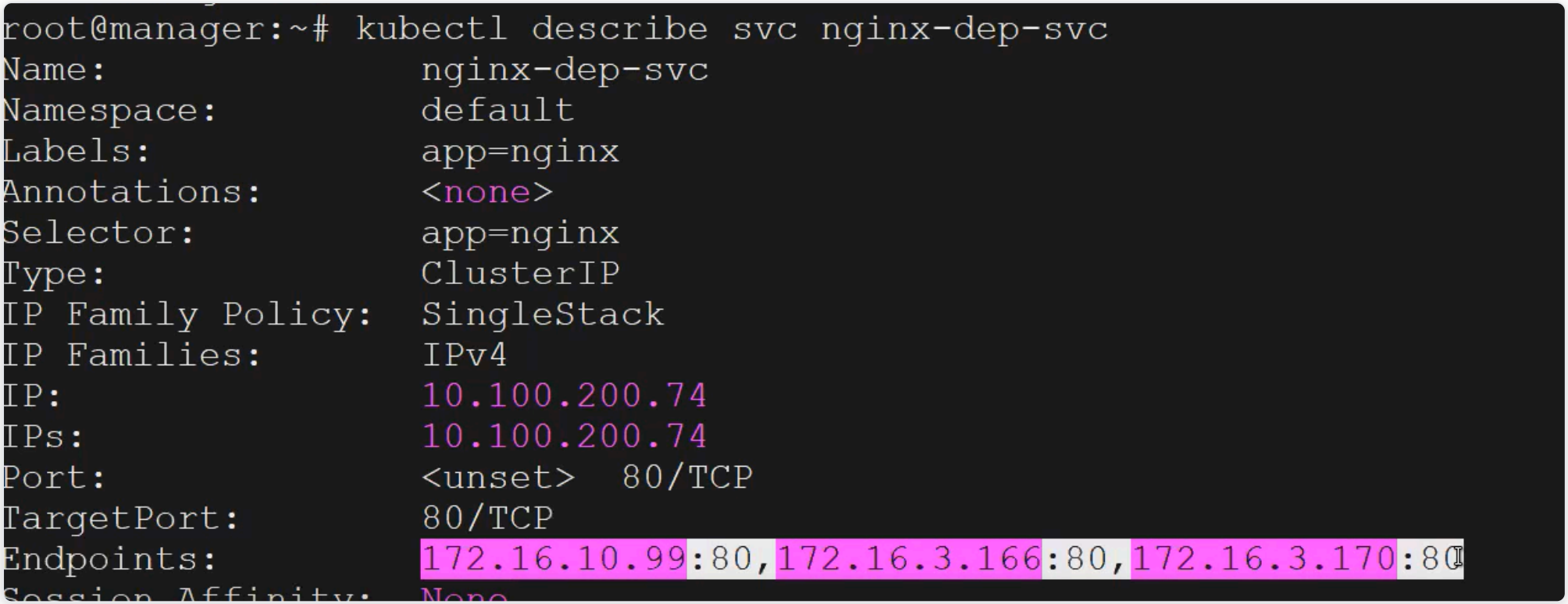

CERTIFIED
kubernetes
ADMINISTRATOR
Deployment
Replica Set
Pod
Pod
Pod
Service
User
Selector of Service
=
Label of Pod


CERTIFIED
kubernetes
ADMINISTRATOR
Deployment
Replica Set
Pod
Pod
Pod
Service
User
Network Nuts - India's Most Loved IT Training Company
www.networknuts.net | info@networknuts.net
Selector of Service
||
Label of Pod


root@manager:~#
root@manager:~#
root@manager:~# kubectl expose deployment nginx-deployment --name nginx-dep-svc

```
root@manager:~#
root@manager:~# kubectl expose deployment nginx-deployment --name nginx-dep-svc
service/nginx-dep-svc exposed
root@manager:~#
root@manager:~# kubectl get svc
NAME             TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)   AGE
kubernetes       ClusterIP   10.96.0.1       <none>        443/TCP   21d
live-svc         ClusterIP   10.111.217.74   <none>        80/TCP    2d4h
nginx-dep-svc    ClusterIP   10.100.200.74   <none>        80/TCP    4s
root@manager:~#
```

```
root@manager:~# kubectl get svc
NAME             TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)   AGE
kubernetes       ClusterIP   10.96.0.1       <none>        443/TCP   21d
live-svc         ClusterIP   10.111.217.74   <none>        80/TCP    2d4h
nginx-dep-svc    ClusterIP   10.100.200.74   <none>        80/TCP    4s
root@manager:~#
```


root@manager:~# kubectl describe svc nginx-dep-svc
Name: nginx-dep-svc
Namespace: default
Labels: app=nginx
Annotations: <none>
Selector: app=nginx
Type: ClusterIP
IP Family Policy: SingleStack
IP Families: IPv4
IP: 10.100.200.74
IPs: 10.100.200.74
Port: <unset> 80/TCP
TargetPort: 80/TCP
Endpoints: 172.16.10.99:80,172.16.3.166:80,172.16.3.170:80

```
IP:               10.100.200.74
IPs:              10.100.200.74
Port:             <unset>  80/TCP
TargetPort:       80/TCP
Endpoints:        172.16.10.99:80,172.16.3.166:80,172.16.3.170:80
Session Affinity:  None
Events:           <none>
root@manager:~# kubectl get pods -l "app=nginx" -o wide
NAME                                READY   STATUS    RESTARTS   AGE   IP              NODE
      NOMINATED NODE   READINESS GATES
nginx-deployment-6fb4f44f88-65kqd   1/1     Running   0          16m   172.16.10.99    node
one   <none>           <none>
nginx-deployment-6fb4f44f88-vbrln   1/1     Running   0          21m   172.16.3.170    node
two   <none>           <none>
nginx-deployment-6fb4f44f88-vdwfv   1/1     Running   0          30m   172.16.3.166    node
two   <none>           <none>
```